J Autism Dev Disord (2008) 38:926–940
DOI 10.1007/s10803-007-0463-4

ORIGINAL PAPER

Effects of the Use of Visual Strategies in Play Groups for Children with Autism Spectrum Disorders and their Peers

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Published online: 12 October 2007
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Abstract The purpose of this study was to investigate the impact of visual strategies with preschool children with autism spectrum disorders (ASD) and their peers during play group sessions. A changing-criterion design was implemented with three preschool-aged children with ASD while they participated in play groups with four typically-developing peers. Results indicated improvements in the use of script phrases, context-related comments, and intervals in which speech occurred for all three participants. Results regarding unscripted phrases, responses, and use of prompts were variable and are discussed.

Keywords Autism · Scripts · Verbal communication · Visual strategies · Unscripted speech

Deficits in social and communication skills are primary characteristics of individuals with autism spectrum disorders (ASD) (Baron-Cohen 2004). In particular, such individuals may not frequently respond to peers or adults, initiate interactions, or speak spontaneously (Janzen 2003; Krantz and McClannahan 1993). The use of visual strategies within the context of heterogeneous play groups (i.e., groups that include both children with disabilities and typically-developing children) is a strategy that may improve such skills in children with ASD.

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Visual information is thought to be more easily processed by individuals with ASD than auditory information (Schopler et al. 1995). This has been demonstrated via higher performance on non-verbal intelligence tests than verbal ones (Scherer et al. 2001) and the presence of hyperlexia, or advanced sight word recognition, in many individuals with ASD (Nation et al. 2006). Further, high-functioning adults with ASD have noted that they respond better to visual information and think visually and are often agitated by auditory input that would not bother most others (Grandin and Scariano 1986; Willey 1999).

Visually-based interventions include visual (e.g., picture, written) schedules, visually organized tasks, written or pictorial scripts, rule reminder cards, and visual task analyses (Simpson et al. 2008). Such interventions may be more effective than auditory-based instruction for individuals with ASD (Bryan and Gast 2000; Simpson et al. 2008). Visual schedules have been found to decrease off-task (Bryan and Gast 2000) and other disruptive behaviors (Dooley et al. 2001), and to improve social skills (Morrison et al. 2002), following directions (Dooley et al. 2001), and task engagement (Bryan and Gast 2000; Dettmer et al. 2000; MacDuff et al. 1993; Massey and Wheeler 2000). Written scripts, with or without pictures, have been reported in a small number of published studies to be effective with individuals with ASD. They have been used to decrease aggressive behavior (Sasso et al. 1990) and to increase bids for attention (Krantz and McClannahan 1998), conversational skills (Charlop-Christy and Kelso 2003), social interactions (Stevenson et al. 2000) and play abilities (Gonzalez-Lopez and Kamps 1997; Johnston et al. 2003; Kamps et al. 1992; Pierce and Schreibman 1995; Pierce and Schreibman 1997; Strain et al. 1979), imitation (Pierce and Schreibman 1995), and other communication skills (Krantz and

McClannahan 1993; Loveland and Tunali 1991; Sarokoff et al. 2001).

Play groups that include both children with ASD and their typically-developing peers may be an effective means to improve play and communication skills in children with ASD, however, only a few empirical studies have been published on this subject. Use of the Integrated Play Group model have resulted in improved and increased social-play skills (Lantz et al. 2004; Wolfberg and Schuler 1993; Yang et al. 2003), functional play with toys (Wolfberg and Schuler 1993), pretend play (Yang et al. 2003; Zurcher et al. 2001), joint attention (Zurcher et al. 2001), verbal interactions (Zurcher et al. 2001), and decreased stereotyped activities and solitary play (Wolfberg and Schuler 1993).

The purpose of this study was to investigate the impact on communication of visual strategies with preschool children with ASD and their peers during play group sessions. The current study contributes to the small literature basis for visually-based interventions and expands on previous research by implementing visual cues within the context of dramatic play activities that include typically-developing peer models, which has not been examined previously. Specific research questions included the following. Would participants' use of script phrases increase in stepwise fashion along with the implementation of scripts and peer instruction cards and would these results generalize to two additional play themes? Would participants' use of unscripted phrases increase over the implementation of scripts and peer instruction cards? Would participants' use of context-related comments increase over the implementation of scripts and peer instruction cards? Would participants' use of responses to peers and adults increase over the implementation of scripts and peer instruction cards? Would participants' use of speech increase over the implementation of scripts and peer instruction cards? Would the levels of prompts used during the implementation of scripts be decreased over the course of implementation and generalization activities?

Method

Participants

Two play groups were formed. Initially, we intended to form two play groups, each with four children, two of whom were to have ASD and two of whom were to be typically-developing. However, we encountered numerous difficulties in recruiting the required number of children, possibly due to the occurrence of this project during the summer months. We received several inquiries from parents of children who were subsequently unable to participate due to the co-occurrence of the play groups and out-of-town trips or other summer activities. Thus, ultimately, the group structures were different than those intended and the two groups differed in that one included one child with ASD, Dario, and two typically-developing peers while the other included two children with ASD, Max and Eddie, and one typically-developing peer.

Criteria for participants with ASD included: (a) must be between 3 and 6 years of age, (b) must have a current diagnosis of autism or pervasive developmental disorder, not otherwise specified (PDD-NOS) given by a medical practitioner (e.g., physician, child psychiatrist) or school diagnostician or school psychologist independently of this research, and (c) must be able to speak in understandable phrases of one or more words. Participants with ASD were recruited through an online group including members of the local chapter of the Autism Society of America. An email was sent to all members of this group describing the project and criteria for participation. Diagnoses were confirmed via a review of the children's school records, which were provided by their mothers. Evaluation records were sparse and included no intelligence testing. The *Gilliam Autism Rating Scale* (*GARS*; Gilliam 1995) was completed by each child's mother prior to the beginning of the study and by the first author following the study to confirm their diagnoses and provide more detailed information regarding the participants' social and communication skills. Two additional participants with ASD were considered, but were unable to participate due to the dates during which the study took place during the summer or schedule conflicts with other services they were receiving.

Dario

Dario was a Hispanic boy who was 4 years, 6 months old at the time of this study. He was currently enrolled in the preschool in which the study took place and had an assistant with him at all times in the classroom he regularly attended. This study took place during the summer. In addition to 30 h of preschool each week, he received occupational therapy and speech therapy each for 45 min per week at school and behavioral therapy at home for approximately 20 h per week, implemented by his mother. During the previous school year, in addition to speech and occupational therapies and behavioral therapy at home, Dario was enrolled in a public school preschool classroom for children with disabilities for 15 h per week.

Dario was diagnosed with high-functioning autism at two-and-a-half-years of age. According to a *GARS* (Gilliam 1995) completed by his mother and one completed by the first author following completion of the study, Dario scored in the "average" range regarding probability of autism. According to the first author's ratings, he scored in the

range of "average" for the stereotyped behaviors and social interaction subtests, and "very high" for the communication subtest regarding probability of autism. In terms of stereotyped behaviors, Dario (a) often avoided eye contact with peers and adults, (b) frequently engaged in visual self-stimulatory behaviors (e.g., repetitively throwing paper and coins in the air to watch them fall, repetitively turning over tubes of liquids to watch them swirl around), and (c) often made repetitive, unintelligible vocalizations. Dario's communication was characterized by (a) frequent repetitions of words and sounds, immediately after hearing them and after delays; (b) unusual, gravelly vocal quality; (c) difficulty responding to one-step directions (e.g., "sit in the chair"); (d) use of single-word utterances; (e) failure to initiate verbal interactions with peers or adults (e.g., requesting items, commenting, conversations); and (f) difficulty answering simple questions (e.g., although he was able to identify colors and numbers, he infrequently did so when asked). Dario's social skills were delayed as well. He (a) did not demonstrate a change in affect when praised, (b) did not imitate peers or adults, (c) played alone and did not play with toys as they were meant to be played with, (d) sometimes tried to leave the play area in avoidance of his peers and adults, (e) occasionally lined objects up, and (f) failed to respond to play initiations made by peers and adults. Dario's mother reported that he appeared to develop normally until approximately 10 months of age when he underwent a catheterization for a bladder and kidney problem. While he was making some consonant and vowel sounds prior to 10 months, afterward, he no longer did. Dario had received no diagnoses other than high-functioning autism. His hearing had been tested and was within normal ranges. Although he had been diagnosed with high-functioning autism, his *GARS* score and behavior in play group indicated that he appeared to have autism, particularly in consideration of his difficulties communicating.

Max

Max was a Caucasian boy who was 4 years, 5 months old at the time of this study. He was not enrolled in the preschool in which the study took place; however, he received 11 h of behavioral therapy per week at a private clinic during the summer. During the previous school year, Max was enrolled in a public school preschool classroom for children with disabilities for 15 h per week, which included approximately 30 min of speech therapy per week, and received 5 h of behavioral therapy per week at a private clinic.

Max was diagnosed with a speech delay at 2 years of age and autism at 3 years of age. According to a *GARS* (Gilliam 1995) completed by his mother, Max scored in the "low" range regarding probability of autism and according to a *GARS* completed by the first author following completion of the study, Max scored in the "below average" range regarding probability of autism. According to the first author's ratings, he scored in the range of "below average" for the stereotyped behaviors and social interaction subtests, and "average" for the communication subtest regarding probability of autism. In terms of stereotyped behaviors, Max often avoided eye contact with peers and adults, frequently walked on tip-toe, and occasionally engaged in repetitive tensing of his body (i.e., seated with legs stiff and straight out in front, arms bent at 90° with hands in fists, and face in a grimace). Max's communication was characterized by (a) occasional repetitions of words and sounds, immediately after hearing them and after delays; (b) some difficulty responding to one-step directions (e.g., "sit by Ash"); (c) use of complete sentences; (d) some difficulty answering questions about books read-aloud; and (e) some difficulty initiating verbal interactions with peers or adults. Though Max frequently made context-appropriate comments, he often "talked to himself" rather than getting the attention of a peer or adult via eye contact, proximity, or saying someone's name. Max's social skills were characterized by (a) occasional withdrawal in group situations, (b) some difficulty imitating peers in play situations, (c) a tendency to "look through" people who spoke directly to him, (d) difficulty adjusting to changed routines and new situations and settings, and (e) occasional tantrums in response to other children taking toys he wanted and adults asking him to do things he did not want to do. Max's mother reported that he demonstrated delays in speech, lack of imitation skills (e.g., waving goodbye), compulsive behaviors and extreme difficulty with transitions, and that he appeared to be deaf when he was under 3 years of age. Max's score on the *GARS* and our observations indicated that Max was high-functioning, particularly considering his ability to communicate effectively in several-word phrases and frequent social initiation.

Eddie

Eddie was a Caucasian boy who was 4 years, 7 months old at the time of this study. He was not enrolled in the preschool in which the study took place; however, he received 30 h of behavioral therapy per week, which included 45–90 min of direct one-on-one instruction per day, at a private behaviorally-based school during the summer and school year. In addition, Eddie received 1 h of private speech therapy and 2 h of private occupational therapy per week and 45 min of private physical therapy per month.

Eddie was diagnosed with strabismus and far-sightedness and was diagnosed with PDD-NOS at 16 months of age. According to a *GARS* (Gilliam 1995) completed by his mother, Eddie scored in the "low" range regarding probability of autism and according to a *GARS* completed by the first author following completion of the study, Eddie scored in the "below average" range regarding probability of autism. According to the first author's ratings, he scored in the range of "very low" for the stereotyped behaviors subtest, and "average" for the communication and social interaction subtests regarding probability of autism. Regarding stereotyped behaviors, Eddie (a) often avoided eye contact with peers and adults, (b) lined up and arranged toys instead of playing with them appropriately, and (c) made high-pitched, repetitive vocalizations occasionally. Eddie's communication was characterized by (a) occasional repetitions of words and sounds, immediately after hearing them and after delays; (b) an unusually high-pitched voice for his age; (c) some difficulty responding to one-step directions; (d) use of complete sentences; (e) a tendency to look away or ignore an adult or peer who spoke to him; (f) difficulty answering questions about books read-aloud; and (g) failure to initiate verbal interactions with peers or adults. Though Eddie occasionally made context-appropriate comments, he often "talked to himself" rather than getting the attention of a peer or adult. Eddie's social skills were characterized by (a) frequent withdrawal and attempts to escape in group situations, (b) lack of imitation of peers or adults in play situations, (c) a tendency to "look through" people who spoke directly to him, (d) difficulty adjusting to changed routines and new situations and settings, and (e) frequent crying in response to other children taking toys he wanted and adults asking him to do things he did not want to do. Eddie's mother reported that he had frequent ear infections and had tubes implanted in his ears and that he appeared to be deaf when he was under 3 years of age, but that his hearing had been tested and was normal. Eddie's score on the *GARS* and our observations indicated that Eddie was high-functioning, particularly considering his ability to communicate effectively in several-word phrases.

Table 1 provides *GARS* autism quotients and subtest scores, according to the first author's ratings, as well as some demographic information for each participant. Developmental subtest scores were taken from each of the participants' mothers' ratings.

Table 1 Participants' *GARS* autism quotients and subtest standard scores and demographic information

Name (pseudonym)	Diagnoses	Age	Race	*GARS* scores and Probability of Autism			
				Autism quotient	Stereotyped behaviors	Communication	Social interaction	Developmental
Dario	High-functioning autism	4 years, 6 months	Hispanic	108 average	9 average	17 very high	10 average	9 average
Max	Autism	4 years, 5 months	Caucasian	87 below average	7 below average	10 average	7 below average	8 average
Eddie	PDD-NOS, strabismus, and far-sightedness	4 years, 7 months	Caucasian	82 below average	3 very low	10 average	9 average	7 below average

Typically-Developing Peers

Four typically-developing peers participated in the play groups as peer models. Each play group consisted of three children. Dario's group consisted of him and two

typically-developing 4-year-old Caucasian girls, Judith and Kyla who were currently enrolled in the preschool in which the play group took place. Max and Eddie's group included one typically-developing child. Geri, a 5-year-old Caucasian girl, participated for the first 2 weeks. Ash, a 4-year-old Caucasian boy, participated for the final 2 weeks. Neither Geri nor Ash was currently attending the preschool, though Ash had attended until a month-and-a-half prior to the study. Preference was given to children currently attending the preschool when choosing typically-developing peers. Criteria for typically-developing participants included: (a) must be between 4 and 5 years old, (b) must not have an identified disability or suspected delays, (c) must be able to speak in intelligible sentences of four or more words, (d) should be children with good attendance and who attend full-time (criterion d applied only to those children attending the preschool but the necessity for good attendance during the study was explained to parents of children who did not previously attend the preschool).

Setting, Play Themes, and Materials

This study took place in a private preschool for typically-developing children, though the preschool did enroll some children with disabilities and developmental delays. Specifically, the play groups took place in a separate, spare classroom, which was connected to a classroom for infants. That is, while they were two separate classrooms, the play group classroom and the infant room were connected by a diaper-changing area and had only a half-wall between them. Therefore, if the infants were crying, it was easily heard in the play group classroom. During baseline and intervention, the children played in an enclosed play area measuring about 55 square feet, surrounded on two sides by child-sized furniture, approximately 2–3 feet tall, and on two sides by walls.

Three play themes were implemented for each group. All of the children participated in an "ocean theme park" theme for intervention. Max and Eddie's play group participated in farm and birthday party themes for generalization. Dario's group participated in farm and bakery themes for generalization. These were chosen with input from the participants' mothers regarding their preferred activities (e.g., the participants lived near an ocean-themed amusement park and had visited) and those with which they were familiar (e.g., one participant frequently engaged in echolalia related to birthday parties). Materials available for the "ocean theme park" included stuffed-animal puppets (i.e., whale, dolphin, fish, crab, turtle, and octopus), a cash register with play money and tickets for the park, a blow-up child's pool, play food, plates, cups, and utensils. Materials for the "farm" included farm-hand clothes (i.e., vest, hat, bandana), stuffed animal puppets (i.e., two ducks, two pigs, two dogs, and two cows), small plastic animals (i.e., cows, horses, pigs, sheep, ducks, chickens, geese, and a dog), and three toy tractors. Materials for the "birthday party" included a wooden birthday cake with decorations (i.e., candles, strawberries, candies) and cake slices that could be attached with hook-and-loop fasteners, party hats, plates, cups, utensils, balloons, and gift bags with tissue paper. Materials for the "bakery" included a basket with a variety of toy breads (e.g., bread sticks, buns, loaves, cupcakes), the wooden birthday cake, toy baking utensils (e.g., spoons, cake pan, pastry cutter, rolling pin, muffin tin, cookie cutters, bowls, spatula), and a toy oven. Max and Eddie's play group participated in all four themes, though data was collected only during the first three and Dario's play group participated in all themes except "birthday party" because he was ill and was absent from preschool for the entire week (week 3) that theme occurred.

The researchers developed scripts to encourage theme-related speech in the children with ASD during each play group session. The scripts that the participants with ASD used during intervention were printed on 2½ by 7½ inch white paper mounted on 3½ by 8½ inch colored paper. The script cards were laminated and each had a 1–2 inch picture (i.e., line drawing or photo) corresponding with the words written on them, which were between 1 and 6 word phrases. In addition, during intervention, the typically-developing peers were introduced to peer instruction cards printed on 8½ by 5½ inch paper, which included pictures (i.e., line drawings or photos) and written instructions on how to interact with their peers with ASD. Scripts and peer instruction cards are described below in the *script instruction* section.

Procedure

The play groups occurred for 4 weeks, for 30 min a day, for 4–5 days each week. Data collection for Max and Eddie occurred during the first 3 weeks of their play group. Dario was ill during the entire third week of his play group, so his data were collected during weeks 1, 2, and 4. Data were collected in 5-min sessions during thematic play. Data collection sessions occurred 1–3 times per day. A changing criterion single-subject design was used to demonstrate collateral changes in the use of scripted phrases in relation to changes in task requirements (Hartmann and Hall 1976). Play themes were introduced approximately one per week; however, all criterion changes were made during the first thematic play unit, "ocean theme park", for all three participants; thus, this theme took slightly longer than a week to complete. Each participant progressed to the next

criterion when he met the current criterion for three consecutive sessions. The criteria for each phase began at one interval (i.e., one script card was implemented within a single interval) and increased by one interval for Dario and Eddie due to the low amounts of speech used during baseline. The criteria for each phase began at two intervals (i.e., two script cards were implemented, each during a separate interval) and increased by two intervals for the second criterion and by one for the third criterion for Max because he spoke more than the other two participants during baseline. All sessions were videotaped to enable data collection.

Visual inspection was used to interpret the graphed results. Additionally, percentage of nonoverlapping data (PND; Scruggs and Mastropieri 1998) were calculated for each participant for each category of data. PND is calculated by dividing the number of intervention data points that are higher than the highest baseline data point by the total number of intervention data points and multiplying the result by 100. According to Scruggs and Mastropieri, PND scores over 90% indicate very effective strategies, scores between 70% and 90% are considered effective, those between 50% and 70% are questionable, and those lower than 50% are not effective.

Baseline

Data were collected for all participants during the "ocean theme park" theme. The children were told they were "here to play" and were shown the materials and were given some suggestions for what they could do with each item (e.g., "the cash register is for buying and selling park tickets"). If any children left the play area, they were returned and told that it was time to play. If any children tried to engage the adults in conversation, they were reminded to talk to and play with their friends and their questions were answered briefly. Geri participated in Max and Eddie's play group during baseline and script intervention.

Script Instruction

Following baseline, the typically-developing peers were told that their friends with ASD would need their help learning to talk and play. Then, within the 15 min prior to intervention and generalization for each new play theme, the typically-developing peers were introduced to one peer instruction card. The peer instruction cards included the following instructions: (a) to get your friend's attention, talk to him and hand him a toy; (b) play with the toys your friend is playing with; and (c) when your friend talks, you say something too. The typically-developing peers were shown a peer instruction card, told what it said, asked to verbally repeat it, and then they role-played the instruction with the adult. While the peer instruction cards were posted on the walls within the play area, beyond initial daily instruction, the peer instruction cards were not pointed out by the adults during play group sessions.

The lengths of the scripts were individualized for each participant based on his verbal and reading abilities. Furthermore, because Max used more context related and unscripted phrases during baseline, he was introduced to twice as many script phrases during the first two criteria. Script instruction for the participants with ASD took place within the 15 min prior to the play group meeting in which each new play theme (i.e., prior to intervention and generalization) was introduced. In addition, immediately prior to each criterion change in intervention, new scripts were introduced. That is, within the 15 min prior to play group, Dario and Eddie were each initially taught one script phrase at a time, then a second script phrase was taught to each participant prior to the increase of the criterion to two intervals, then a third script phrase was taught immediately prior to the increase to the criterion to three intervals. Within the 15 min immediately prior to each generalization play theme, they were each taught the three script phrases relevant to those themes. Max's instruction followed the same pattern, however, he was first taught 2 phrases, then two more (criterion = 4 intervals), then one more (criterion = 5 intervals). Each script was practiced with each participant at least five times or until he was able to recite it independently upon being shown the script. Samples of script phrases for each participant for each theme are illustrated in Table 2.

Script Intervention

Each participant with ASD participated in the "ocean theme park" play theme for intervention. Procedures in place during baseline continued. In addition, during each 5-min session, typically-developing peers were reminded of the directions on the peer instruction cards. Script phrases were introduced during intervention based on the current criterion level for each participant (e.g., when the criterion was one phrase/interval, the script card was shown to the participant at least once during that session). Criteria were increased when the participant had met the current level for three consecutive sessions. No more than one script phrase was held in front of a participant per 20-s interval; however, the script cards were placed in the play area near relevant materials. If participants were already verbally interacting with peers or spontaneously and independently recited script phrases, these interactions were not interrupted. Each script card was used once per session and no more than one script card was prompted during an interval. Scripts were shown to the participants

Table 2 Samples of script phrases for each participant

	Theme 1 (Ocean theme park)	Theme 2 (Farm)	Theme 3 (Dario: Bakery; Max and Eddie: Birthday)
Dario	A fish!	A duck.	Rolls, please.
Max	Look at the dolphins.	The horse is running.	Let's decorate the cake.
Eddie	Do you see the purple fish?	"Neigh, neigh," says the horse.	Blow out the candles.

according to the following prompt sequence/levels, waiting 5 s between each step:

1. Held script card up in front of participant, about 2–3 feet away from face,
2. Hand-over-hand helped participant point to the card,
3. Verbally modeled the words written on the card,
4. Told the participant to, "Say _____ [what the card says]."

Prompts were delivered by the first and second authors and a trained undergraduate research assistant. Prompting stopped when the participant recited the script phrase. If the participant did not respond after all of the steps were completed, the instructor waited at least one interval before trying again. Whenever possible, the scripts were shown to the participants when they were already in close proximity or playing with the related materials.

Generalization

Procedures implemented during intervention were the same as those implemented during generalization, however, prompting decreased and occurred at less intrusive levels. The final criterion for intervention was used to determine how many script phrases to implement in the following two generalization play themes. The participants were taught all of the script phrases for each new theme at one time, unlike during intervention, when participants learned only one or two phrases until mastery of that criterion level. Dario's play group participated in the "farm" and "bakery" during generalization. Max and Eddie's play group participated in the "farm" and "birthday party" during generalization. Ash's participation began during the "farm" play theme.

Procedural Integrity

Procedural integrity was observed based on the prompt sequence described above in the *script intervention* section. Instructors were rated overall for the session to determine if the steps were implemented correctly and if the instructors waited 5 s between each step. Procedural integrity was calculated by dividing the number of steps implemented correctly by the total possible steps completed. Twenty-seven percent of Dario's intervention and generalization sessions were observed for procedural integrity, which was 100% for each session. Twenty-five percent of Max's intervention and generalization sessions were observed for procedural integrity, which was 100% for each session. Twenty-seven percent of Eddie's intervention and generalization sessions were observed for procedural integrity, which was 100% for each session.

Response Definitions

Data were collected via interval recording. Intervals were each 20 s long and there were 15 intervals per session. Each interval was observed to determine whether or not the participant used *script phrases* and/or *unscripted phrases*. Both scripted and unscripted phrases could be recorded as occurring during a single interval; however, the frequency of these statements was not recorded. Only the occurrence or non-occurrence of each within the interval was recorded. *Script phrases* were recorded as occurring when an interval included words, phrases, or sentences that were word-for-word the same as the phrases listed on the participants' written scripts or within one word of the written scripts. Any other intelligible words, phrases, or sentences were recorded as *unscripted phrases*. In addition, intervals were recorded as including *context-related comments* whenever a *script phrase* was used and when an *unscripted phrase* used during an interval related to the current theme, an action performed by a peer or adult, or something said by a peer or adult during the play group.

Intervals were also recorded as including *responses* when the participant answered a peer's or adult's question within 5 s or immediately imitated a peer's or adult's speech. Percentages of intervals in which each of the above response definitions occurred were calculated for each session. In addition, the percentage of *intervals in which speech occurred* was calculated for each session and included all intervals in which any of the above response definitions occurred. That is, each session was analyzed overall for the occurrence or non-occurrence of any verbal communication and this was calculated as a percentage of the total possible intervals (i.e., an interval that contained the coding of the occurrence of any of the above response definitions was coded as an occurrence of speech). Finally,

the level of prompt required was recorded each time a participant used a *script phrase*, according to the definitions of each prompt level given above in the *script intervention* section.

Inter-Observer Reliability

Inter-observer reliability was collected by the first and second authors. Both have terminal degrees and experience in conducting and collecting data on single-subject research. Reliability was calculated by dividing the number of agreements by the total number of agreements plus disagreements, multiplied by 100 (Kazdin 1982). An agreement was scored when both observers independently recorded the same code in an interval (e.g., script phrase [S], unscripted phrase [U], context-related comment [C], responses [R]). Reliability was assessed during baseline sessions for 33% of Dario's, Max's, and Eddie's sessions and was assessed at 95%, 92%, and 89% agreement respectively. Reliability was assessed during intervention and generalization sessions for 33% of Dario's, 29% of Max's, and 33% of Eddie's sessions and was assessed at mean agreements of 98%, 89%, and 89% respectively. Agreement was not assessed on prompting level.

Results

Script Phrases

Figure 1 illustrates each participant's results for script phrases. A functional relation was demonstrated between the intervention and the participants' behaviors. There was a successive increase in scripted verbalizations and all but one data point met or exceeded the established criterion for all participants. Dario's use of script phrases averaged 0% of the intervals observed during baseline, and met or exceeded criterion levels for each intervention and generalization session. His use of script phrases during intervention averaged 7% (range = 7%) during the first criterion level (7%), 27% (range = 13–40%) during the second criterion level (13%), 25% (range = 20–27%) during the third criterion level (20%), 27% (range = 27%) during the first generalization play theme (20%), and 20% (range = 20%) during the second generalization play theme (20%). Max's use of script phrases averaged 0% of the intervals observed during baseline, and met or exceeded criterion levels for all intervention and generalization sessions except one. His use of script phrases during intervention averaged 22% (range = 20–27%) during the first criterion level (13%), 29% (range = 20–33%) during the second criterion level (27%), 33% (range = 33%) during the third criterion level (33%), 35% (range = 33–40%) during the first generalization play theme (33%), and 44% (range = 33–53%) during the second generalization play theme (33%). Eddie's use of script phrases during intervention averaged 0% of the intervals observed during baseline, and met or exceeded criterion levels for each session. His use of script phrases averaged 7% (range = 7%) during the first criterion level (7%), 20% (range = 13–27%) during the second criterion level (13%), 25% (range = 20–27%) during the third criterion level (20%), 31% (range = 27–40%) during the first generalization play theme (20%), and 22% (range = 20–27%) during the second generalization play theme (20%). There was no overlap between baseline and intervention data points for any of the participants, thus, the PND was 100%, indicating a very effective treatment.

Unscripted Phrases

Figure 1 illustrates each participant's results for unscripted phrases. Dario's use of unscripted phrases averaged 9% (range = 0–13%) of the intervals observed during baseline. His use of unscripted phrases during intervention averaged 2% (range = 0–7%) during the first criterion level, 5% (range = 0–7%) during the second criterion level, 7% (range = 0–13%) during the third criterion level, 2% (range = 0–7%) during the first generalization play theme, and 4% (range = 0–13%) during the second generalization play theme. Max's use of unscripted phrases averaged 40% (range = 33–47%) of the intervals observed during baseline. His use of unscripted phrases during intervention averaged 89% (range = 87–93%) during the first criterion level, 71% (range = 60–80%) during the second criterion level, 53% (range = 47–60%) during the third criterion level, 49% (range = 13–80%) during the first generalization play theme, and 60% (range = 40–73%) during the second generalization play theme. Eddie's use of unscripted phrases averaged 24% (range = 13–33%) of the intervals observed during baseline. His use of unscripted phrases during intervention averaged 16% (range = 7–20%) during the first criterion level, 9% (range = 0–27%) during the second criterion level, 27% (range = 0–40%) during the third criterion level, 40% (range = 33–47%) during the first generalization play theme, and 29% (range = 7–40%) during the second generalization play theme.

To supplement the visual analysis, PND calculated for Dario reveals a score of 0% for unscripted phrases, a score of 82% for Max's unscripted phrases, and a score of 22% for Eddie's unscripted phrases. The PND for unscripted phrases suggests an ineffective treatment for Dario and Eddie and an effective treatment for Max.

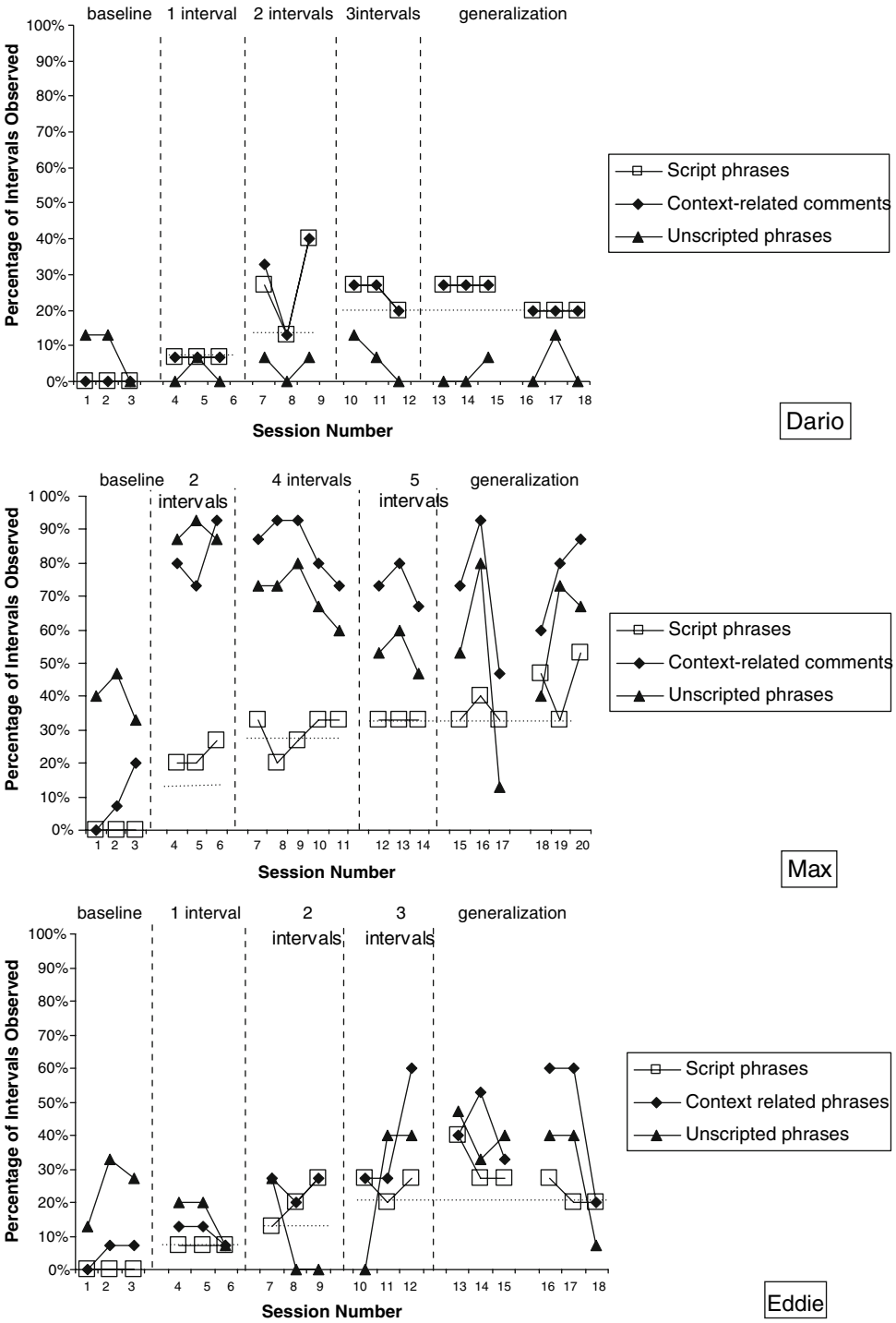


Fig. 1 Script phrases, context-related comments, and unscripted phrases

Context-Related Comments

Figure 1 illustrates each participant's results for context-related comments. A functional relation was demonstrated between the intervention and context-related comments across all participants. There were only 2 overlapping data points (6%) between baseline and intervention. There was a successive increase in the use of context-related comments across all participants. Dario's use of context-related comments averaged 0% of the intervals observed during baseline. His use of context-related comments during intervention averaged 7% (range = 7%) during the first criterion level, 29% (range = 13–40%) during the second criterion level, 25% (range = 20–27%) during the third criterion level, 27% (range = 27%) during the first generalization play theme, and 20% (range = 20%) during the second generalization play

theme. Max's use of context-related comments averaged 9% (range = 0–20%) of the intervals observed during baseline. His use of context-related comments during intervention averaged 82% (range = 73–93%) during the first criterion level, 85% (range = 80–93%) during the second criterion level, 73% (range = 67–80%) during the third criterion level, 71% (range = 47–93%) during the first generalization play theme, and 76% (range = 60–87%) during the second generalization play theme. Eddie's use of context-related comments averaged 5% (range = 0–7%) of the intervals observed during baseline. His use of context-related comments during intervention averaged 11% (range = 7–13%) during the first criterion level, 25% (range = 20–27%) during the second criterion level, 38% (range = 27–60%) during the third criterion level, 42% (range = 33–53%) during the first generalization play theme, and 47% (range = 20–60%) during the second generalization play theme.

To supplement the visual analysis, PND calculated for Dario reveals a score of 100% for context-related phrases, a score of 100% for Max's context-related phrases, and a score of 89% for Eddie's context-related phrases. The PND for context-related phrases suggests a very effective treatment for Dario and Max and an effective treatment for Eddie.

Responses

Figure 2 illustrates each participant's results for responses. Dario's use of responses averaged 2% (range = 0–7%) of the intervals observed during baseline. His use of responses during intervention averaged 0% during the first criterion level, 0% during the second criterion level, 7% (range = 0–13%) during the third criterion level, 0% during the first generalization play theme, and 2% (range = 0–7%) during the second generalization play theme. Max's use of responses averaged 4% (range = 0–13%) of the intervals observed during baseline. His use of responses during intervention averaged 22% (range = 13–33%) during the first criterion level, 26% (range = 13–40%) during the second criterion level, 22% (range = 7–40%) during the third criterion level, 20% (range = 20%) during the first generalization play theme, and 11% (range = 0–20%) during the second generalization play theme. Eddie's use of responses averaged 4% (range = 0–13%) of the intervals observed during baseline. His use of responses during intervention averaged 4% (range = 0–13%) during the first criterion level, 0% during the second criterion level, 25% (range = 0–47%) during the third criterion level, 22% (range = 7–47%) during the first generalization play theme, and 11% (range = 0–27%) during the second generalization play theme.

To supplement the visual analysis, PND calculated for Dario reveals a score of 0% for responses, a score of 73% for Max's responses, and a score of 22% for Eddie's responses. The PND for responses suggests an ineffective treatment for Dario and Eddie and an effective treatment for Max.

Intervals in which Any Speech Occurred

Figure 2 illustrates each participant's results for intervals with any speech. Intervals in which Dario used any speech averaged 11% (range = 0–20%) of the intervals observed during baseline. His use of speech during intervention averaged 9% (range = 7–13%) during the first criterion level, 29% (range = 13–40%) during the second criterion level, 31% (range = 20–40%) during the third criterion level, 27% (range = 20–33%) during the first generalization play theme, and 24% (range = 20–33%) during the second generalization play theme. Intervals in which Max used any speech averaged 42% (range = 40–47%) of the intervals observed during baseline. His use of speech during intervention averaged 95% (range = 93–100%) during the first criterion level, 89% (range = 80–93%) during the second criterion level, 78% (range = 73–80%) during the third criterion level, 75% (range = 60–93%) during the first generalization play theme, and 82% (range = 60–93%) during the second generalization play theme. Intervals in which Eddie used any speech averaged 24% (range = 13–33%) of the intervals observed during baseline. His use of speech during intervention averaged 24% (range = 13–33%) during the first criterion level, 33% (range = 20–53%) during the second criterion level, 62% (range = 27–80%) during the third criterion level, 71% (range = 67–73%) during the first generalization play theme, and 53% (range = 27–73%) during the second generalization play theme.

To supplement the visual analysis, PND calculated for Dario reveals a score of 44% for use of any speech, a score of 100% for Max's use of any speech, and a score of 33% for Eddie's use of any speech. The PND for use of any speech suggests an ineffective treatment for Dario and Eddie and a very effective treatment for Max.

Prompts

Figure 3 illustrates the highest and lowest levels of prompts given to each participant in each session. Dario's prompts ranged from level 1 to level 4 during the first criterion level for intervention, from level 0 to level 4 in the second criterion level, from 0 to 3 in the third criterion level, from 0 to 3 in the first generalization play theme, and from 2 to 4 in the final generalization theme. Max's

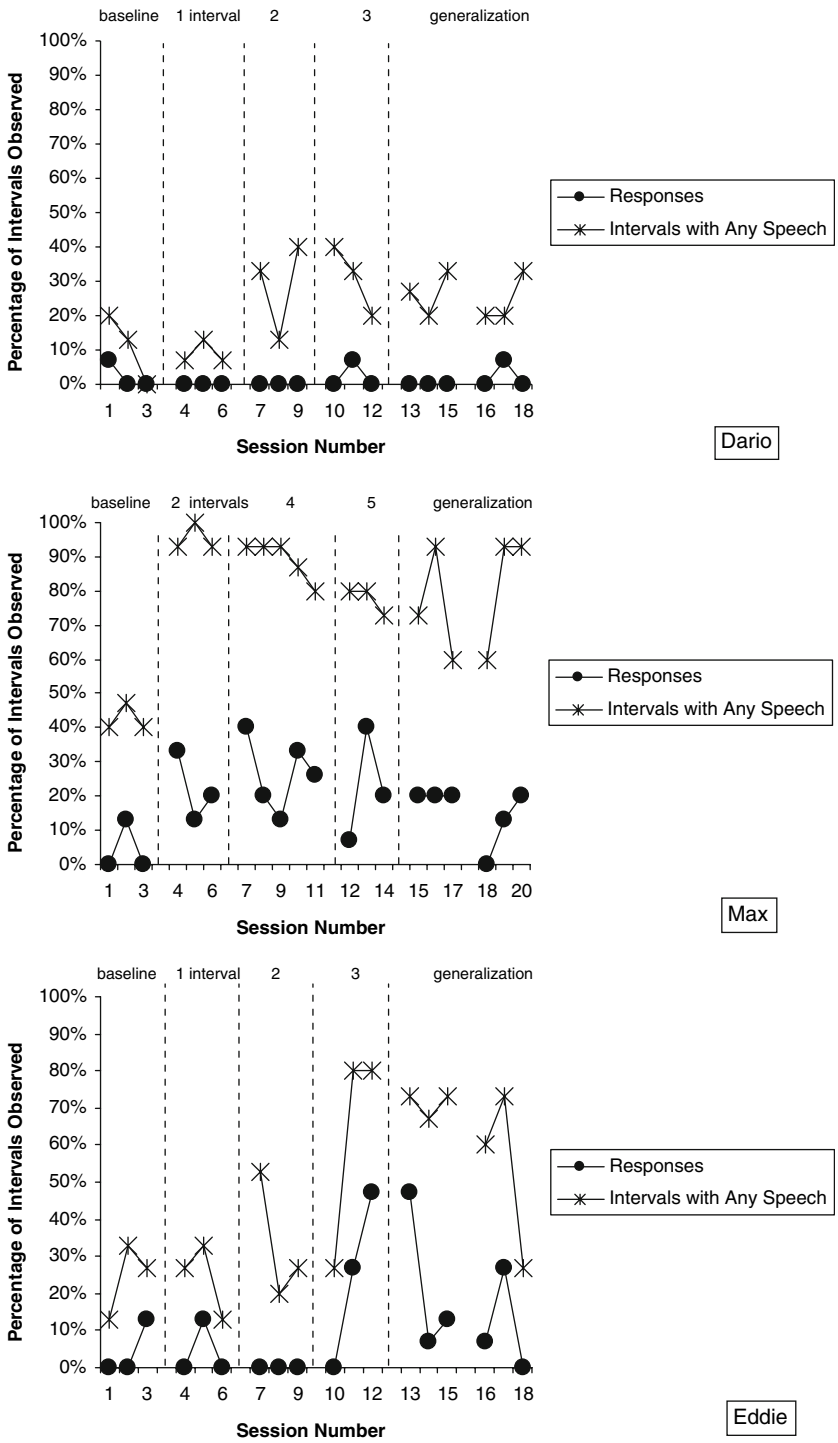


Fig. 2 Responses and intervals with any speech

prompts ranged from level 0 to level 2 during the first criterion level for intervention, from level 0 to level 4 in the second criterion level, from 1 to 3 in the third criterion level, from 1 to 3 in the first generalization play theme, and from 0 to 3 in the final generalization theme. Eddie's prompts were all at level 1 during the first criterion level for intervention, from level 1 to level 4 in the second criterion level, from 1 to 4 in the third criterion level, from 1 to 3 in the first generalization play theme, and from 1 to 4 in the final generalization theme.

Discussion

Consistent with the research literature in the area of heterogeneous play groups, the participants' play behaviors

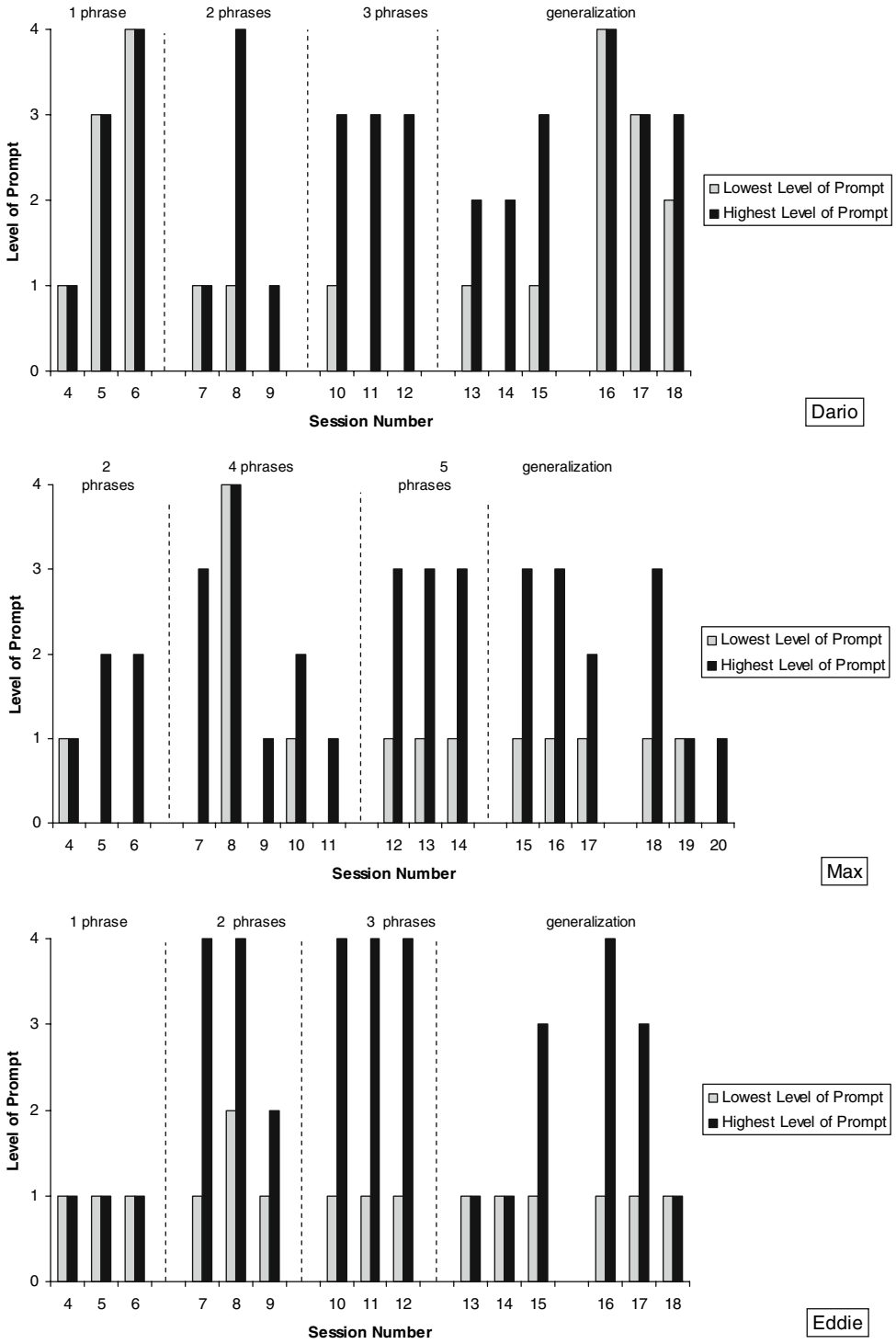


Fig. 3 Levels of prompts

increased with regard to their context-related language (Lantz et al. 2004; Wolfberg and Schuler 1993; Yang et al. 2003; Zurcher et al. 2001). This study extended the line of research on heterogeneous play groups by including a visually-based language intervention. A functional relation was demonstrated between the visually-based intervention and play-related language across play themes as well as for all participants. This is consistent with previous research in which visually-based interventions led to increases in communication, language, and social skills (Charlop-Christy and Kelso 2003; Gonzalez-Lopez and Kamps 1997; Johnston et al. 2003; Kamps et. al. 1992; Krantz and McClannahan 1998; Krantz and McClannahan 1993; Loveland and Tunali 1991; Pierce and Schreibman 1997; Pierce and Schreibman 1995; Sasso et al. 1990; Sarokoff et al. 2001; Stevenson et al. 2000; Strain et al. 1979). This

study extended the lines of research in the areas of integrated play and visually-based interventions by demonstrating increases in language by implementing components of the two interventions.

Script phrases systematically increased for all three participants as the script intervention was implemented. PND results for scripted phrases for all three participants indicate that this is a very effective intervention. Also importantly, each participant occasionally exceeded criterion levels during intervention phases. This occurred at times when participants demonstrated immediate echolalia that overlapped use of the same script phrase into following intervals, at time due to delayed echolalia occurring within later intervals in the same session, or when participants independently picked up script cards and recited their scripts. Independent use of scripts in preschool children with ASD is encouraging, particularly when considering the viability of the use of this strategy with older individuals with ASD and the possibility of using scripts inconspicuously with peers, rather than relying on prompts and cues from adults. Results regarding script phrases were maintained during all generalization sessions.

Unscripted phrases remained at low rates from baseline through intervention and generalization for Dario, increased from baseline to intervention then slightly decreased during generalization for Max, and remained approximately the same for Eddie. Further, PND results for unscripted phrases for two participants indicate that this is an ineffective intervention and indicate that it was an ineffective intervention for one participant, Max. However, changes in the amount of context-appropriate comments may be more important to consider in terms of the abilities of children with ASD to "fit in" in play situations with their peers. PND results for context-related phrases for the three participants indicate that this is a very effective or effective intervention. In particular, though Max and Eddie exhibited moderate rates of unscripted phrases during baseline, those statements were rarely related to the play theme, actions of peers or adults in the room, or related to what peers or adults in the play area were talking about. Context-related comments increased steadily from baseline to intervention and to generalization for Dario, closely following his progress in script phrases. Context-related comments were also low for Max and Eddie during baseline and increased during intervention and generalization. While Max and Eddie's context-related comments were not as closely matched with script phrases as were Dario's, there is a clear relationship between an increase in script phrases and an increase in context-related comments. In fact, Max's context-related comments, script phrases, and unscripted phrases all increased dramatically following intervention implementation, demonstrating that the speech he used during intervention and generalization was of higher quantity and quality than the comments he made during baseline.

Responses to peers and adults were recorded for each participant to determine if the peer instruction cards used with the typically-developing peers resulted in increased responding from the children with ASD. PND results for responses indicate that this is not an effective intervention for two of the participants, Dario and Eddie. Dario, who spoke the least of the three participants with ASD, only responded three times, once during baseline, once during the final criterion level of intervention, and once during generalization. Dario's responses included saying, "no," and echoing words said by peers or adults. Max did demonstrate a modest increase in responses from baseline to intervention and generalization and Eddie demonstrated a modest increase in responses in the final criterion increase of intervention and during generalization. This may be due to the difference in the makeup of their play group. Because they had only one typically-developing peer, that child was more likely to interact with his or her playmates with ASD. In addition, Max frequently used context-related comments during intervention and generalization, which may have encouraged Geri and Ash to interact with him. Similarly, Eddie spoke more and made more context-related comments during the final criterion level and during generalization, which also may have encouraged more initiations from Geri and Ash. It is also possible that the difference in makeup of the two groups may have impacted the participants' performance. Placing two children with ASD in the same group provides fewer socially-competent peer models than placing one child with ASD in a group with two socially-competent models; however, in this example, Max and Eddie were high-functioning and Max, in particular, initiated communication with his peer with ASD and typically-developing peers.

Intervals in which any speech occurred increased for all participants throughout intervention and generalization; however, PND results for use of any speech indicate that this is a very effective intervention for only one participant, Max, and an ineffective treatment for the other participants. Dario's speech increased during the second criterion level of intervention and those levels were maintained during the third criterion level and generalization. Max's speech increased and was maintained throughout intervention and generalization, however, observers noted that while he spoke more often than the other two participants with ASD, he did not always speak to anyone in initial sessions of intervention, instead speaking to himself, though this improved, particularly during generalization. Eddie's speech increased during the final criterion level and was maintained through generalization.

Ideally, prompts would have been decreased to level 0 (self-prompt) or 1 (held script card in front of participant)

in all sessions by the end of intervention. This did not occur for any of the participants with ASD. However, both Dario and Max did, at times, self-prompt by picking up the script phrase cards and reciting the scripts (sessions 9, 11, 12, and 14 for Dario and sessions 5–7, 9, 11, and 20 for Max). While Eddie did not self-prompt with the scripts, he usually required no more than a level 1 prompt. It is possible that with a longer intervention or with older participants, prompts may be decreased to all level 0 or 1 prompts.

This study was limited in that it only involved three children with ASD and four typically-developing peers. In addition, results may only be generalized to children of a similar age and similar communication skills. Dario's peers, Judith and Kyla, initiated toward him infrequently, which may have limited his opportunities to communicate. This may have been due to two factors, the makeup of his play group and his communication abilities. Judith and Kyla were close friends prior to this play group and easily interacted with each other, but did not easily include Dario, possibly due to his infrequent use of speech, difficulty understanding what he did say, and his infrequent attempts to interact with them, despite visual and verbal reminders each session regarding their jobs listed on the peer instruction cards.

Future research may expand this intervention for use with older individuals with ASD (e.g., elementary-aged, high school students, adults) and to individuals with less verbal ability or more. Future research may also investigate how to best implement integrated play groups between preschoolers with ASD and typically-developing peers, particularly how to increase the initiations of typically-developing children to their peers with ASD (e.g., via self-monitoring procedures or reinforcement), providing training on autism to desensitize typically-developing peers to unfamiliar characteristics and behaviors, and on what types of visual strategies are most effective with typically-developing peers to increase their initiations toward peers with ASD. Further, one-on-one pairs that include one child with ASD and one typically-developing child may be most effective and efficient in increasing interactions. As interactions with the child with ASD and a typically-developing peer increase, perhaps the number of play group members could be increased successively to resemble more a typical play situation found in inclusive settings. A longer implementation of visual scripts could lead to decreased prompts and independent use of scripts. Further, future studies may investigate the use of generic phrases that could be used in multiple contexts. Finally, investigations of the use of visual cues may have wide application in a variety of contexts in which individuals with ASD are integrated with their peers.

Acknowledgments This study was funded by a grant from the College of Education and Human Development at the University of Texas at San Antonio. The authors wish to express their gratitude to the parents of the participants for allowing us to play with their children and to the Director, Alissa Levey Baugh, and staff at the Block and Dreeben School for Young Children for allowing us to implement the project on their site and for their ongoing support throughout the project and to Heidi Hawbecker who assisted in video recording and implementation of the play groups.

References

Baron-Cohen, S. (2004). Autism: Research into causes and intervention. *Pediatric Rehabilitation, 7*, 73–78.

Bryan, L. C., & Gast, D. L. (2000). Teaching on-task and on-schedule behaviors to high-functioning children with autism via picture activity schedules. *Journal of Autism & Developmental Disorders, 30*, 553–567.

Charlop-Christy, M. H., & Kelso, S. E. (2003). Teaching children with autism conversational speech using a cue card/written script program. *Education and Treatment of Children, 26*, 108–127.

Dettmer, S., Simpson, R. L., Myles, B. S., & Ganz, J. B. (2000). The use of visual supports to facilitate transitions of students with autism. *Focus on Autism & Other Developmental Disabilities, 15*, 163–169.

Dooley, P., Wilczenski, F. L., & Torem, C. (2001). Using an activity schedule to smooth school transitions. *Journal of Positive Behavior Interventions, 3*, 57–61.

Gilliam, J. E. (1995). *Gilliam autism rating scale*. Austin: PRO-ED, Inc.

Gonzalez-Lopez, A., & Kamps, D. (1997). Social skills training to increase social interactions between children with autism and their typical peers. *Focus on Autism and Other Developmental Disabilities, 12*, 12–14.

Grandin, T., & Scariano, M. M. (1986). *Emergence: Labeled autistic*. New York: Warner Books.

Hartmann, D., & Hall, R. V. (1976). The changing criterion design. *Journal of Applied Behavior Analysis, 9*, 527–532.

Janzen, J. E. (2003). *Understanding the nature of autism: A guide to the autism spectrum disorders* (2nd ed.). San Antonio: Therapy Skill Builders.

Johnston, S., Nelson, C., Evans, J., & Palazolo, K. (2003). The use of visual supports in teaching young children with autism spectrum disorder to initiate interactions. *Augmentative and Alternative Communication, 19*, 86–103.

Kamps, D., Leonard, B., Vernon, S., Dugan, E., Delquadri, J., Gerson, B., Wade, L., & Folk, L. (1992). Teaching social skills to students with autism to increase peer interactions in an integrated first grade classroom. *Journal of Applied Behavior Analysis, 25*, 281–288.

Kazdin, A. E. (1982). *Single-case research designs: Methods for clinical and applied settings*. New York: Oxford University Press.

Krantz, P., & McClannahan, L. (1993). Teaching children with autism to initiate to peers: Effects of a script-fading procedure. *Journal of Applied Behavior Analysis, 26*, 121–132.

Krantz, P. J., & McClannahan, L. E., (1998). Social interaction skills for children with autism: A script-fading procedure for beginning readers. *Journal of Applied Behavior Analysis, 31*, 191–202.

Lantz, J. F., Nelson, J. M., & Loftin, R. I. (2004). Guiding children with autism in play: Applying the integrated play group model in school settings. *TEACHING Exceptional Children, 31*, 8–14.

Loveland, K., & Tunali, B. (1991). Social scripts for conversational interactions in autism and Down's syndrome. *Journal of Autism and Developmental Disorders, 21*, 177–186.

MacDuff, G. S., Krantz, P. J., & McClannahan, L. E. (1993). Teaching children with autism to use photographic activity schedules: Maintenance and generalization of complex response chains. *Journal of Applied Behavior Analysis, 26*, 89–97.

Massey, N. G., & Wheeler, J. J. (2000). Acquisition and generalization of activity schedules and their effects on task engagement in a young child with autism in an inclusive pre-school classroom. *Education and Training in Mental Retardation and Developmental Disabilities, 35*, 326–335.

Morrison, R. S., Sainato, D. M., BenChaaban, D., & Endo, S. (2002). Increasing play skills of children with autism using activity schedules and correspondence training. *Journal of Early Intervention, 25*, 58–72.

Nation, K., Clarke, P., Wright, B., & Williams, C. (2006). Patterns of reading ability in children with autism spectrum disorder. *Journal of Autism and Developmental Disorders, 36*, 911–919.

Pierce, K., & Schreibman, L. (1995). Increasing complex social behaviors in children with autism: Effects of peer-implemented pivotal response training. *Journal of Applied Behavior Analysis, 28*, 285–295.

Pierce, K., & Schreibman, L. (1997). Multiple peer use of pivotal response training to increase social behaviors of classmates with autism: Results from trained and untrained peers. *Journal of Applied Behavior Analysis, 30*, 157–160.

Sarokoff, R. A., Taylor, B. A., & Poulson, C. L., (2001). Teaching children with autism to engage in conversational exchanges: Script fading with embedded textual stimuli. *Journal of Applied Behavior Analysis, 34*, 81–84.

Sasso, G. M., Melloy, K. J., & Kavale, K. (1990). Generalization, maintenance, and behavioral co-variation associated with social skills training through structured learning. *Behavioral Disorders, 16*, 9–22.

Scherer, M., Pierce, K. L., Paredes, S., Kisacky, K. L., Ingersoll, B., & Schreibman, L. (2001). Enhancing conversational skills in children with autism via video technology. Which is better, 'Self' or 'Other' as a model? *Behavior Modification, 25*, 140–158.

Schopler, E., Mesibov, G. B., & Hearsey, K. (1995). Structured teaching in the TEACCH system. In E. Schopler & G. B. Mesibov (Eds.), *Learning and cognition in autism* (pp. 243–267). New York: Plenum.

Scruggs, T. E., & Mastropieri, M. A. (1998). Summarizing single-subject research: Issues and applications. *Behavior Modification, 22*, 221–242.

Simpson, R. L., Myles, B. S., & Ganz, J. B. (2008). Efficacious interventions and treatments for learners with autism spectrum disorders. In R. L. Simpson & B. S. Myles (Eds.), *Educating children and youth with autism: Strategies for effective practice* (2nd ed., pp. 477–512). Austin, TX: Pro-Ed, Inc.

Stevenson, C. L., Krantz, P. J., & McClannahan, L. E. (2000). Social interaction skills for children with autism: A script-fading procedure for nonreaders. *Behavioral Interventions, 15*, 1–20.

Strain, P., Kerr, M., & Ragland, E. (1979). Effects of peer-mediated social initiations and prompting/reinforcement procedures on the social behavior of autistic children. *Journal of Autism and Developmental Disorders, 9*, 41–54.

Willey, L. H. (1999). *Pretending to be normal: Living with Asperger's syndrome*. London: Jessica Kingsley Publishers.

Wolfberg, P. J., & Schuler, A. L. (1993). Integrated play groups: A model for promoting the social and cognitive dimensions of play in children with autism. *Journal of Autism and Developmental Disorders, 23*, 467–489.

Yang, T., Wolfberg, P. J., Wu, S., & Hwu, P. (2003). Supporting children on the autism spectrum in peer play at home and school. *Autism, 7*, 437–453.

Zurcher, C., Hunt, P., Schuler, A., Webster, J. (2001). Increasing joint attention, play and language through peer-supported play. *Autism, 5*, 374–398.